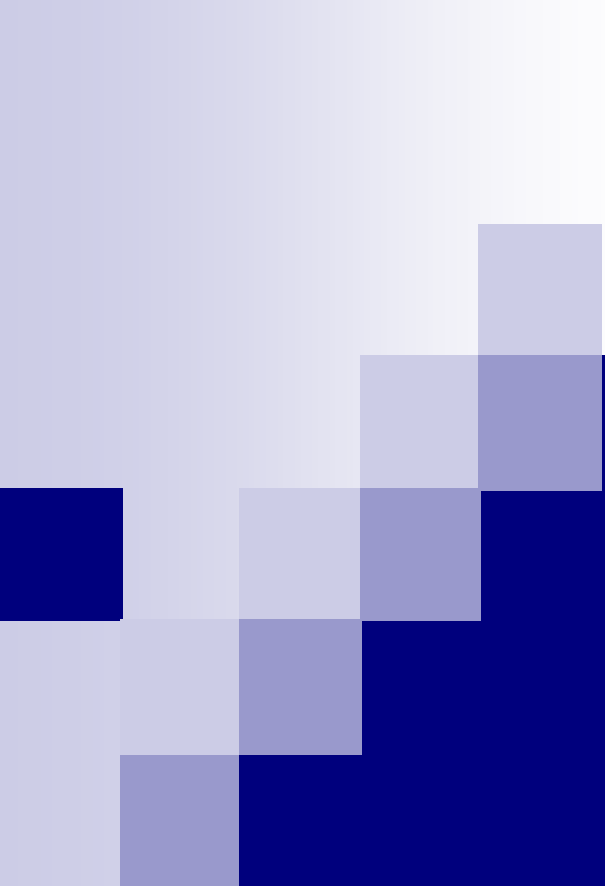




SQA Reviews

Lecture # 14

What is a Review?

- A process or meeting during which a work product, or a set of work products, is presented to project personnel, managers, users, or other interested parties for comment or approval. Types include code review, design review, formal qualification review, requirements review, test readiness review
 - IEEE Std. 610.12-1990

Objectives of Reviews - 1

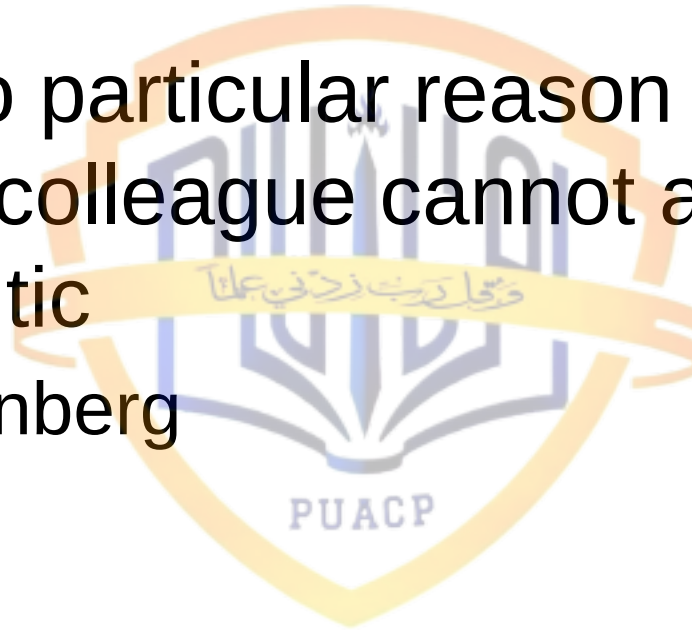
- Identify required improvements in a product
- Assure that the deliverable is complete
- Assure that the deliverable is technically correct
- Measure the progress of the project

Objectives of Reviews - 2

- Identify any defects early, thus resulting in cost and time savings
- Assure the quality of deliverable before the development process is allowed to continue
- Once a deliverable has been reviewed, revised as necessary, and approved, it can be safely used as a basis for further development

Colleagues as Critics

- There is no particular reason why your friend and colleague cannot also be your sternest critic
 - Jerry Weinberg



Benefits of Review

- A number of team members get an opportunity to provide their input
- Ownership of the work product is transferred from an individual to a group
- A (limited) training ground

Kinds of Reviews

- Business reviews
- Technical reviews
- Management reviews
- Walk-throughs
- Inspections



Objectives of Business Reviews

- The deliverable is complete
- The deliverable provides the information required for the next phase
- The deliverable is correct
- There is adherence to the procedures and policies

Objectives of Technical Reviews - 1

- Point out needed improvements in the product of a single person or a team
- Confirm those parts of a product in which improvement is either not desired or not needed
- Achieve technical work or more uniform, or at least more predictable, quality than can be achieved without reviews, in order to make technical work more management

Objectives of Technical Reviews - 2

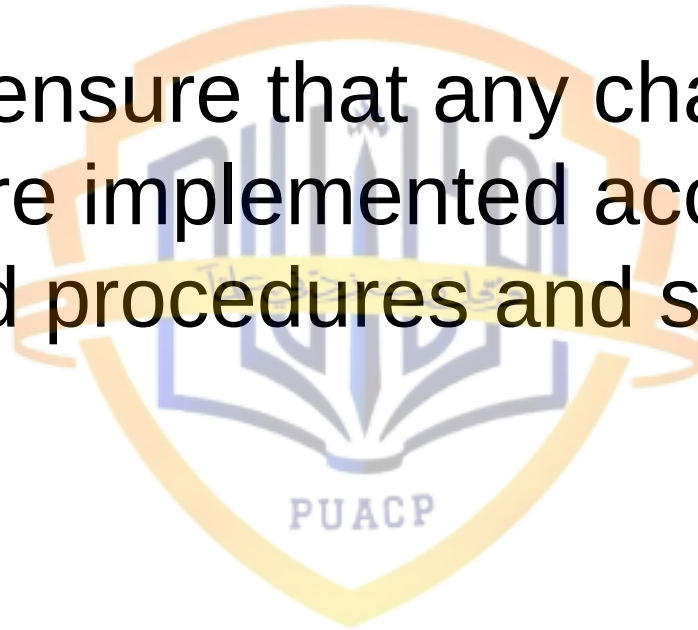
- Software reviews are a “filter” for software engineering process
- Reviews are applied at several points during software development and serve to uncover errors and defects that can then be removed
- Software reviews “purify” the software engineering activities

Objectives of Technical Reviews - 3

- Technical work needs reviewing for the same reason that pencils need erasers: To err is human
- Another reason we need technical reviews is that although people are good at catching some of their own errors, large classes of errors escape the originator more easily than they escape anyone else

Objectives of Technical Reviews - 4

- They also ensure that any changes to the software are implemented according to pre-defined procedures and standards



What Technical Reviews Are Not!

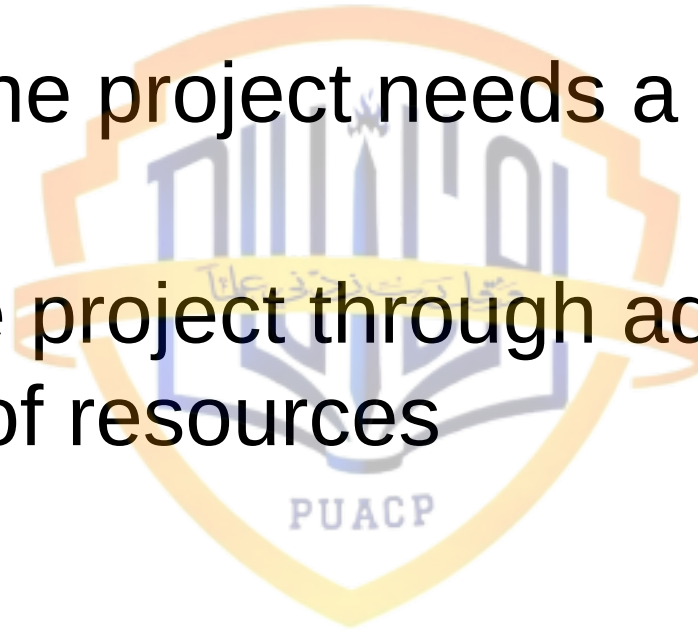
- A project budget summary
- A scheduling assessment
- An overall progress report
- A mechanism for reprisal or political intrigue!!

Objectives of Management Reviews - 1

- Validate from a management perspective that the project is making progress according to the project plan
- Ensure a deliverable is ready for management approval
- Resolve issues that require management's attention

Objectives of Management Reviews - 2

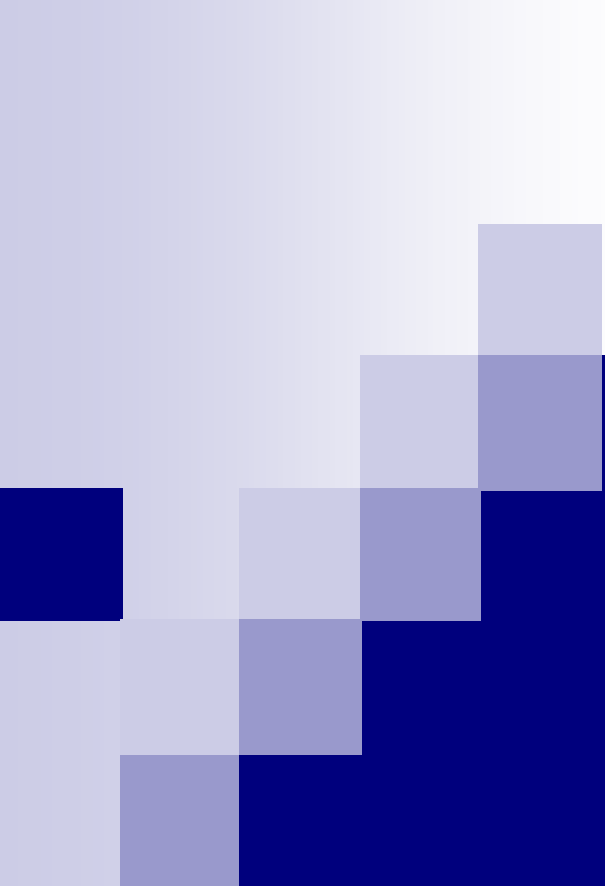
- Identify if the project needs a change of direction
- Control the project through adequate allocation of resources



Review Roles

- Facilitator
- Author
- Recorder
- Reviewer
- Observer





Responsibilities of Roles



Responsibilities of Facilitator

- Responsible for providing the background of the work and assigning roles to attendees
- Encourages all attendees to participate
- Keeps the meeting focused and moving
- Responsible for gaining consensus on problems

Responsibilities of Author - 1

- Responsible for the readiness and distribution of material to be reviewed
- During the meeting, the author paraphrases the document a section at a time
- Responsible for
 - scheduling the review
 - selecting the review participants
 - determining if the entry criteria for the review are met

Responsibilities of Author - 2

- providing information about the product during all stages
- clarifying any unclear issues
- correcting any problems identified
- providing dates for rework and resolution

Responsibilities of Recorder

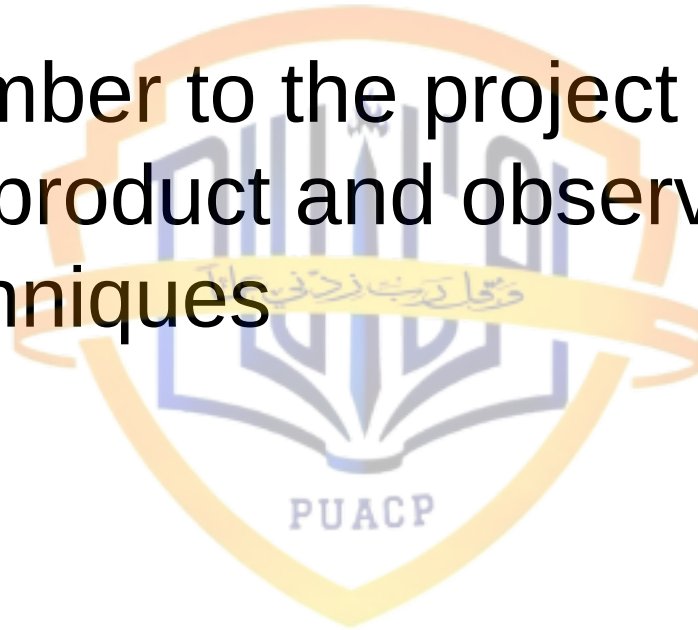
- Collects and records each defect uncovered during the review meeting
- Develops an issues list and identifies whose responsibility it is to resolve each issue
- Records meeting decisions on issues; prepares the minutes; and publishes the minutes, and continually tracks the action items

Responsibilities of Reviewer

- Spends time prior to the meeting reviewing information
- Makes notes of defects and becomes familiar with the product to be reviewed
- Identifies strengths of the product
- Verifies that the rework is done
- Insists upon clarifying any issues that are not clear

Responsibilities of Observer

- A new member to the project team, who learns the product and observes the review techniques



Review Guidelines

- Preparation
- Discussions
- Respect
- Agenda
- Review Records
- Resources
- Attendees



Review Frequency

- At the beginning/end of the requirements phase
- At the beginning/end of the design phase
- At the beginning/end of the code phase
- At the beginning/end of the test phase
- Approval of the test plan

Review Planning - 1

- Distribute review package one week in advance
 - Document to be reviewed
 - Review agenda
 - Identification of the individual who will manage the agenda and schedule
 - Exit and entrance criteria for the review
 - Objective of the review

Review Planning - 2

- Names of attendees, their roles and responsibilities
- Review location
- Date and time of review
- List of classifications that will be used for defects discovered (defect type, defect origin, and defect severity)
- Procedures for handling issues raised during the review and escalation phase

Review Meeting - 1

- Facilitator begins the meeting with an introduction of agenda, people, and description of their roles
- Author of the document proceeds to explain the materials, while reviewers raise issues based on advance preparation

Review Meeting - 2

- When valid problems, issues, or defects are discovered, they are classified according to their origin or severity and then recorded
- These are accompanied with the names of individuals who are responsible for resolution and the time frame during which the item will be resolved
- Related recommendations are also recorded

Guidelines for Reviewers

- Be prepared - evaluate product before the review meeting
- Review the product, not the producer
- Keep your tone mild, ask questions instead of making accusations
- Stick to the review agenda
- Raise issues, don't resolve them
- Avoid discussions of style - stick to technical correctness

Decisions at the End of a Review Meeting

- All attendees must decide whether to
 - Accept the product without further modification
 - Reject the product due to severe errors
 - Accept the product provisionally
 - Hold a follow-up review session

Review Report - 1

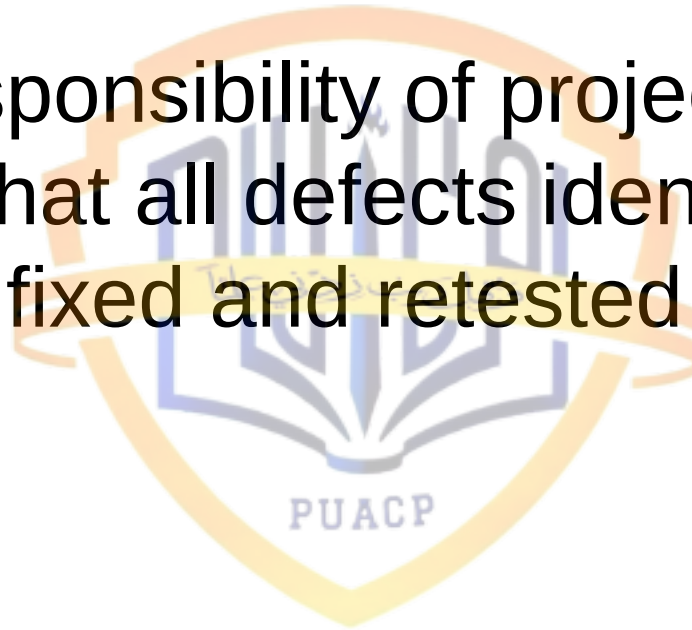
- Published by the recorder, with approval from all attendees, after a week of the review meeting
- Review report consists of
 - Elements reviewed
 - Names of individuals who participated in the review
 - Specific inputs to the review

Review Report - 2

- ☐ List of unresolved items
- ☐ List of issues that need to be escalated to management
- ☐ Action items/ownership/status
- ☐ Suggested recommendations

Rework

- It is the responsibility of project manager to ensure that all defects identified in the review are fixed and retested



Follow-Up

- During the follow-up, that all discrepancies identified are resolved and the exit criteria for the review have been met
- Document lessons learned during the final report also

Summary

- Discussed different kinds of reviews: business, technical, and management
- Introduced the review process, meeting, and post-review process

References

- Inroads to Software Quality by Alka Jarvis and Vern Crandall, PH 1997 (Ch. 7)
- Software Engineering: A Practitioner's Approach by Roger S. Pressman (Ch. 8)